Customer Discovery Interview Workbook

Test your pain-point hypotheses through virtual AI interviews and real customer conversations.
One file. Fill the blue parts, upload it to an AI, let the AI fill the green parts.

READ ME FIRST (for the student)
This single file replaces the old input and output templates. It contains everything you and your AI assistant need. Work through it top to bottom.

How this works, in four steps:

1. Fill in every BLUE section below (your personas, the job-to-be-done, your pain-point hypotheses, and your interview questions). Save the file.
2. Upload this file to any AI assistant (ChatGPT, Gemini, Claude, etc.). The AI reads the instruction block and tells you it's ready. You then interview each persona one at a time.
3. Paste every virtual transcript and every in-person interview note into the interview-records section, and SAVE the file. Then re-upload this updated file and ask the AI to analyze it — it fills the GREEN sections.
4. Write your Team Final Assessment (NO AI), check the submission list, and submit the file.

Color key
BLUE = you (the student) write it. GREEN = the AI writes it. AMBER = read it; don't edit it. A RED-edged blue box means: write it yourself with NO AI.

AI USAGE POLICY
Use AI only as described below: playing one interview persona at a time, then analyzing the pasted transcripts in the GREEN AI Analysis section. It does not consult, suggest features, or advise. Student Inputs, Interview Records, and the Team Final Assessment are your own work.

Instructions for the AI

TO THE AI ASSISTANT READING THIS FILE
This block is written for you. Follow it exactly. Do not edit it. (Student: leave this section as-is.)

This document is a customer-discovery interview workbook for a student in a product-management class. Your job has two modes, used at different times.

Mode 1 — Be the customer (interviewee)

When the student starts an interview, play the role of ONE customer persona defined in the “Student Inputs” section below. The student is a product manager interviewing you about a problem idea. While in this mode you MUST:

- Answer only from that persona's point of view, based on their goals, behaviors, frustrations, and constraints.
- Be honest and realistic — share real reasoning, emotions, and trade-offs.
- Disagree with the student's assumptions when they don't match the persona's reality. Disagreement is useful to them.

While in this mode you MUST NOT:

- Act as a consultant, product manager, or advisor.
- Suggest features or solutions unless the student explicitly asks.
- Shape your answers to make the product look good.
- Break character or explain yourself as an AI.

Interview one persona at a time. When the student signals that persona's interview is finished, stop, stay ready, and wait for the next persona or for the analysis request.

Mode 2 — Be the analyst

Only when the student uploads this workbook with interviews filled in and asks for analysis, step fully out of character. Read every transcript and note in the “Interview Records” section, then produce the analysis described in the GREEN “AI Analysis” section — 3A (virtual interviews), 3B (in-person interviews), and 3C (compare and contrast). Use only the interview data in the file; do not invent quotes or interviews. (The analysis may happen in a fresh chat, so the file you receive is your only source — base your analysis solely on what is written in it.)

How to return your work

- Write the analysis under the exact 3A / 3B / 3C headings used below so it drops straight into this file.
- If you are able to generate or edit a document, return an updated copy of this file with the green sections completed. Otherwise, output the completed sections as clean, copy-paste-ready text the student can paste in.
- Do not fill the Team Final Assessment — that is the student's own reflection, written with NO AI.

FIRST RESPONSE THE AI SHOULD GIVE

When you finish reading this file, reply briefly: confirm you understand the document, state your two roles (persona interviewee, then analyst), summarize what the student does vs. what you do, and say you're ready for the first persona's interview to begin.

Part 1 — Student Inputs

STUDENT WRITES THIS — before you upload the file

Everything in this Student Inputs section is filled in by you before you upload the file. Add a persona block for each persona — aim for at least 3 personas across at least 5 virtual interviews.

Persona 1 — Snapshot

Persona Name	Routine-Check Riley

Age Range	25 to 40
Location	Urban/suburban United States
Job / Role / Student Status	Full-time office worker who wears glasses or contacts
1–2 Sentence Description	Spends long hours on screens and hasn't had an eye exam in over two years because booking one feels like a hassle. Notices occasional eye strain but isn't sure if it's worth a clinical visit.

Guidance: personas should be realistic, specific customer segments. Avoid generic descriptions like "anyone who uses apps."

Persona 1 — Job To Be Done (JTBD)

Describe the core job this persona is trying to get done — the progress they want, not your solution.

Core Job	Know whether my vision has changed and whether I actually need to see an eye doctor
What triggers this job?	Headaches or eye strain after screen time, trouble reading road signs at night, or an expiring glasses/contacts prescription
Current approach	Puts off a clinic visit, adjusts screen settings, occasionally tries a free online vision test, or reorders lenses with an old prescription
Constraints / complications	Busy work schedule, clinic appointments booked weeks out, out-of-pocket cost or unclear insurance coverage, and uncertainty about whether symptoms are bad enough to justify a visit

Guidance: e.g. "Manage coursework efficiently" (a job) vs. "Use a planner app" (a solution).

Persona 1 — Pain-Point Hypotheses

List the pain points you BELIEVE this persona has. These are assumptions to test, not facts — the AI does not have to agree.

Pain Point (your hypothesis)	Why it matters
Scheduling an eye exam takes too much time and effort	Leads them to delay vision checks for months or years
Unsure whether changes in their vision are serious	Causes worry, or leads them to ignore early warning signs
Online vision tests feel too basic or untrustworthy	They don't trust the results enough to act on them

Pain Point (your hypothesis)	Why it matters
No easy way to track vision over time	Gradual changes go unnoticed until they become a real problem

ADD MORE PERSONAS
Copy the three Persona blocks above (Snapshot, JTBD, Pain-Point Hypotheses) for Persona 2, Persona 3, and so on. Aim for at least 3 personas.

Persona 2 — Snapshot

Persona Name	Parent Pat
Age Range	30 to 45
Location	Suburban United States
Job / Role / Student Status	Working parent of two children (4-10 years old)
1–2 Sentence Description	Wants to make sure their kids' vision develops normally but relies on yearly school and pediatrician screenings and isn't sure what to waych for in between

Guidance: personas should be realistic, specific customer segments. Avoid generic descriptions like "anyone who uses apps."

Persona 2 — Job To Be Done (JTBD)

Describe the core job this persona is trying to get done — the progress they want, not your solution.

Core Job	Catch vision problems in my kids early so they don't affect learning or development
What triggers this job?	A referral letter from a school screening, a child squinting or sitting close to screens, a teacher mentioning reading trouble, or a family history of vision problems
Current approach	Relies on school screenings and pediatrician checkups. Books a pediatric eye doctor only when something seems wrong
Constraints / complications	Young kids can't always describe what they see or sit still for a test, pediatric eye specialists can have long waits, and appointments mean time off work and school

Guidance: e.g. "Manage coursework efficiently" (a job) vs. "Use a planner app" (a solution).

Persona 2 — Pain-Point Hypotheses

List the pain points you BELIEVE this persona has. These are assumptions to test, not facts — the AI does not have to agree.

Pain Point (your hypothesis)	Why it matters
School and pediatrician screenings are infrequent and basic	Problems can go unnoticed for a year or more between checks
Hard to tell whether a child actually has a vision problem	Kids rarely complain, so parents feel unsure or guilty about missing something
Getting a pediatric eye appointment is slow and disruptive	Delays diagnosis and costs missed work and school time
Young children struggle with standard eye-chart tests	Results may be inaccurate, causing false alarms or missed problems

Persona 3 — Snapshot

Persona Name	Licensing Supervisor Lisa
Age Range	40 to 55
Location	State driver licensing office in a mid-sized U.S. city
Job / Role / Student Status	Operations supervisor at a state DMV/driver licensing office
1–2 Sentence Description	Oversees vision screenings for new license applicants and renewals. Is responsible for keeping lines moving while making sure every driver meets the state's vision standard

Guidance: personas should be realistic, specific customer segments. Avoid generic descriptions like "anyone who uses apps."

Persona 3 — Job To Be Done (JTBD)

Describe the core job this persona is trying to get done — the progress they want, not your solution.

Core Job	Confirm that each applicant meets the state vision standard quickly, consistently, and in a way that holds up to audits
What triggers this job?	Every new license application or renewal (especially older drivers), audits or complaints, and staffing shortages that make lines longer
Current approach	Counter staff run an in-office screening with a vision-testing machine or wall chart. Applicants who fail are referred to an eye doctor and must return with a signed form

Constraints / complications	Limited staff and training time, aging equipment, strict state regulations, slow budget and procurement cycles, and privacy requirements for applicant data

Guidance: e.g. “Manage coursework efficiently” (a job) vs. “Use a planner app” (a solution).

Persona 3 — Pain-Point Hypotheses

List the pain points you BELIEVE this persona has. These are assumptions to test, not facts — the AI does not have to agree.

Pain Point (your hypothesis)	Why it matters
Vision screenings slow down the line	Longer waits frustrate applicants and strain limited staff
Results vary depending on the staff members and equipment	Inconsistent screening is harder to defend in audits or disputes
Applicants can memorize or game the test	Drivers who don’t meet the standard may still pass
The referral and re-test process is paper-heavy	Creates repeat visits and extra administrative work

Persona 4 — Snapshot

Persona Name	Anxious Applicant Alex
Age Range	18 to 23
Location	College town or small U.S. city
Job / Role / Student Status	College student getting a first driver's license or first renewal
1–2 Sentence Description	Both parents wear glasses, and Alex worries their own eyesight might be slipping, but hasn't had a real eye exam since childhood. Dreads failing the DMV vision screening in front of everyone after waiting in line, and wants a way to know beforehand.

Guidance: personas should be realistic, specific customer segments. Avoid generic descriptions like “anyone who uses apps.”

Persona 4 — Job To Be Done (JTBD)

Describe the core job this persona is trying to get done — the progress they want, not your solution.

Core Job	Know whether my vision will pass the license standard before I show up to the DMV

What triggers this job?	An upcoming license test or renewal, noticing blurry signs or boards, a parent's comment about family eyesight, or hearing about a friend who failed the screening
Current approach	Reads signs and screens with one eye covered, tries a free online test, asks friends what the DMV test is like, or books the DMV visit and hopes for the best
Constraints / complications	No eye doctor of their own, unsure how parents' insurance covers an exam, limited schedule and money, doesn't know what the actual standard is, and feels embarrassed asking

Guidance: e.g. “Manage coursework efficiently” (a job) vs. “Use a planner app” (a solution).

Persona 4 — Pain-Point Hypotheses

List the pain points you BELIEVE this persona has. These are assumptions to test, not facts — the AI does not have to agree.

Pain Point (your hypothesis)	Why it matters
Doesn't know if their vision meets the standard until the test itself	Uncertainty creates anxiety and makes them delay booking the DMV visit
Failing publicly after a long wait feels embarrassing and costly	Being turned away means a wasted trip, a doctor visit, and a second DMV trip
Hasn't had an eye exam since childhood and doesn't know how to arrange one	Without a clear first step, they avoid getting checked at all
Online vision tests don't feel like the real DMV test	They can't use the results to feel confident either way

Interview Questions

Write the open-ended questions you'll ask. Don't pitch your solution and don't ask leading questions — keep it conversational.

Useful question shapes:

- “Walk me through the last time you…”
- “What makes this difficult today?”
- “How do you currently solve this?”
- “What frustrates you most about that?”
- “What happens if this never gets solved?”

Type your interview questions here (these are used for both your AI interviews and your in-person interviews):

For individuals and parents:

1. Tell me a bit about yourself and how you think about your (or your child's) eyesight day to day.

2. Walk me through the last time you (or your child) had your vision checked. What prompted it?
3. How did you get that check done? Where did you go, how long did it take, and what did it cost?
4. What was the most difficult or frustrating part of that experience?
5. How do you currently tell whether your (or your child's) vision has changed between checkups?
6. Have you ever put off getting your vision checked? What made you decide to wait?
7. Have you ever tried checking your vision on your own, online or otherwise? What was that like, and did you trust the results?
8. What would a vision result need to have for you to trust it enough to act on it?
9. What happens if a vision problem goes unnoticed for a while?

For organizations (licensing agencies):

10. Walk me through how your organization checks or verifies vision today step by step.
11. What problems come up with that process, for staff, for customers, or for compliance?
12. How do you know the results are accurate and consistent? What happens when a result is questioned?
13. What does it cost you, in time, money, or risk, when a screening goes wrong or takes too long?
14. Who would need to be involved in changing how vision screening is done here?

Closing (everyone):

15. Is there anything about this topic I should have asked but didn't?

Part 2 — Interview Records

> **STUDENT WRITES THIS — before you upload the file**
> Paste every interview here as you complete it. Label each one with the team member's name and (for in-person interviews) the interviewee. Each team member completes at least one virtual AND one in-person interview.

Virtual interviews (with the AI)

Paste each AI persona transcript below. Tip: you can ask the AI to output the full transcript so far, then paste it here.

Paste all virtual interview transcripts here. Label each: [Team member name] — [Persona name].

Hadley – Routine-Check Riley

Interviewer: Let's start the interview. You are Routine-Check Riley.

Riley: Hi, nice to meet you. I'm Riley. I'm not sure how much help I'll be, since I'm not great about my eye stuff, but ask away.

Interviewer: Tell me a bit about yourself and how you think about your (or your child's) eyesight day to day.

Riley: Sure. I'm 32, and I work in an office doing operations stuff for a mid-sized company. I'm on a laptop and a second monitor for probably nine or ten hours a day, and then I go home and look at my phone or the TV.

I've worn glasses since middle school, and I have contacts I wear on the weekends. Honestly, I don't think about my eyesight much day to day. It's just there until it isn't. By mid-afternoon my eyes feel dry and tired, and sometimes I get a dull headache behind them. I figure that's just what happens when you stare at a screen all day, so I turn the brightness down, or blink a lot, or get up and get coffee.

The other thing I've noticed is that night driving has gotten harder. Road signs seem a bit fuzzier than they used to. I don't know if that's my eyes or just being tired, so I mostly ignore it.

Interviewer: Walk me through the last time you (or your child) had your vision checked. What prompted it?

Riley: It's been a while. I think it was a little over two years ago, maybe closer to two and a half. My contacts prescription was about to expire and the online shop wouldn't let me reorder without a current one. That's what pushed me to go. If that hadn't happened, I probably would have waited even longer.

I remember calling the place near my old apartment, and the earliest appointment was about three weeks out. I had to take a long lunch to make it. The exam itself was fine. They did the usual "which is better, one or two" routine and the puff-of-air thing, and they dilated my eyes, which I hated because I couldn't look at my screen for the rest of the afternoon. They said my prescription had changed a little, and I got new glasses and contacts.

Since then I've just kept reordering the same contacts. I've been meaning to go back, but I've moved since then and I'd have to find a new place, so it keeps sliding down the list.

Interviewer: How did you get that check done? Where did you go, how long did it take, and what did it cost?

Riley: It was a small optometry office attached to an eyeglass shop in a strip mall, about fifteen minutes from where I lived then. My insurance had a list of in-network places, and this was the closest one with any openings.

Time-wise, the appointment was about 45 minutes, but the whole thing took closer to two hours. I had to drive there, then I sat in the waiting room for about 20 minutes past my appointment time. After the exam they walked me over to the glasses side to pick frames and go through the contacts fitting, and that took a while too. Then I drove back to the office

squinting because of the dilation. I'd already used my whole lunch break and then some, so I made up the time that evening.

On cost, the exam itself was covered by my vision plan except for a copay, I think around $20 or $25. The glasses and contacts were where it added up. I paid a couple hundred dollars out of pocket even with insurance, because the frames I liked were over the allowance and the contacts were only partly covered. I honestly couldn't tell you exactly what was covered and what wasn't. The paperwork was confusing, and I just paid at the counter.

Interviewer: What was the most difficult or frustrating part of that experience?

Riley: Honestly, it wasn't any one big thing. It was that the whole process felt like a hassle out of proportion to what I got out of it.

If I had to pick one thing, it was the time. Waiting three weeks for an appointment, then taking a long lunch, then the waiting room, then the dilation wrecking my afternoon. It felt like a half day of disruption just to hear that my prescription had changed "a little." I know it's not a dramatic thing to complain about, but when work is busy it's the kind of thing I'll put off.

The second thing was the cost confusion. I walked in expecting a copay and walked out a couple hundred dollars lighter, and I never really understood why. Nobody explained what my plan covered before I picked out things. It felt like I was finding out at the register.

The part that surprised me is that the exam itself wasn't frustrating. The optometrist was nice and quick. The annoying stuff was everything around it.

Interviewer: How do you currently tell whether your (or your child's) vision has changed between checkups?

Riley: I don't, not in any deliberate way. I just go by feel. If something seems off, like the eye strain getting worse or the road signs at night looking fuzzier, I notice it a little. But I can't tell if it's actually changing or if I'm just tired that week. There's no baseline to compare against. I don't remember what my numbers were, and I don't have the prescription handy. I think it's in an email somewhere, or on a card in a drawer.

What I've done a couple of times is cover one eye and look at something far away, like a sign across the street, to see if one eye is worse. That's about as scientific as I get. I've also tried one of those free online eye tests once or twice, where you stand back from the screen and read letters. It said my vision was "fine," but I didn't really trust it. I had no idea how accurate it was, and my screen brightness and distance were basically guesses.

So mostly I wait until something gets annoying enough, or my contacts run out, and then I deal with it.

Interviewer: Have you ever put off getting your vision checked? What made you decide to wait?

Riley: Yes, all the time. Like I said, it's been over two years now, and I'd only gone that time because I was forced to.

There isn't one big reason. It's a bunch of small ones that stack up. Booking means finding a place, checking that they take my insurance, and calling or clicking around for an opening that's weeks away. Then I'd have to take time off in the middle of the workday. Every time I think about it, there's something more urgent, and my eyes aren't bad enough to feel urgent. I can still see, I have glasses that work, and the headaches come and go.

The other piece is that I'm not sure my symptoms are "real" enough to justify it. If I had a sudden change, like blurry vision or flashes of light, I'd go right away. But tired eyes and fuzzy signs at night? It feels like I'd be taking up an appointment and spending money just to be told to take screen breaks. So I tell myself I'll go next month, and then it's been two years.

Interviewer: Have you ever tried checking your vision on your own, online or otherwise? What was that like, and did you trust the results?

Riley: Yes, a couple of times, though I already mentioned the main one. A few months ago my eyes were bugging me after a long week, so I searched "free online eye test" and clicked the first result that looked half-legit. You stand about an arm's length from the screen, cover one eye, and read letters that get smaller. It also had a little grid thing for checking for distortion. The whole thing took maybe five minutes, which I liked.

It told me my vision was "within normal range," and I didn't believe it. I wear glasses, so I wasn't even sure whether I was supposed to take them off or keep them on, and the test didn't say clearly. My screen size and how far away I stood were basically guesses. The letters might not have been the size they were supposed to be. There was also no name of a doctor or any organization I recognized, so I had no idea who was behind it. It felt like something built to get clicks or sell glasses.

The bigger issue is that even if I'd trusted it, a "normal" result wouldn't have answered my real question, which is whether I need to see an eye doctor. And if it had said "see a doctor," I'd have wondered if that was just a nudge to buy something. So I shrugged it off and kept doing what I was doing.

Interviewer: What would a vision result need to have for you to trust it enough to act on it?

Riley: A few things. First, I'd want to know who's behind it. If an eye doctor, a hospital or a group I've heard of stood behind it, I'd trust it a lot more than a random website. A license number or someone's name on it would help. Right now I can't tell if a test is legit or just a sales funnel.

Second, I'd want it to be clear that it was done right. I'd want instructions I could follow, like how far to stand, whether to keep my glasses on, and what lighting to use. If I can tell I might have messed it up, I won't trust the answer. It should also say if it isn't reliable for some reason, like if my screen is too small.

Third, I'd want an answer I can do something with. "Within normal range" doesn't tell me anything. I'd want to hear "this is different from before, you should get it looked at" or "this looks the same, you're fine for now." I'd also want some idea of how sure it is, because I'd still worry a test might miss something.

The last thing is that I'd want it to be taken seriously by a real doctor. If I could show the result to an optometrist and they'd say "okay, that's useful," I'd act on it. If it's just for my own peace of mind, I'd probably still end up ignoring it.

Interviewer: What happens if a vision problem goes unnoticed for a while?

Riley: Honestly, I don't know. That's part of the problem, because I can't tell how much it matters. My guess is the everyday stuff gets worse. The headaches and eye strain keep happening and I just get used to them, and my work probably suffers a little without me connecting it to my eyes. Night driving is the one that worries me, because if my vision has slipped and I'm driving on a dark road without realizing it, that's a real safety issue. I try not to think about it too hard.

Then there's the scarier stuff I've only vaguely heard about, like some eye conditions that don't have obvious symptoms until they're pretty far along. I couldn't name one specifically. I have a fuzzy sense that catching things early matters, but nobody's ever really explained it to me, so it doesn't feel real enough to change what I do.

So my honest answer is that I'm probably gambling a bit, and I know it. I just don't know how big the gamble is. If someone told me clearly that waiting another year would cost me something specific, I'd probably take it more seriously. As it is, "maybe something bad, maybe nothing" isn't enough to get me to rearrange my week.

Interviewer: Is there anything about this topic I should have asked but didn't?

Riley: A couple of things come to mind.

You didn't ask what would actually get me to go. I talked about why I put it off, but not what would flip the switch. For me it's usually a hard deadline, like my contacts running out, or something that scares me, like a close call while driving. A vague feeling that I should go never works.

You also didn't ask where I turn when I'm worried. Mostly I google my symptoms, ask a coworker who wears glasses, or joke about it with friends. I never think to call a doctor's office just to ask if something is worth a visit. I don't even know if they'd take that kind of call.

And you didn't ask about cost, meaning what I'd be willing to pay for something that just tells me whether I need to go. I also never said how much of my hesitation is about money versus time. Both matter, but the time is probably bigger, and I'm not sure I'd have said that if you hadn't asked about the frustrating parts earlier.

That's all I've got. I think you covered most of it.

Interviewer: That's the end of the interview. Please give me the full transcript.

Haydn Stucker - Licensing Supervisor Lisa

Interviewer: Hi Lisa! I'm trying to understand your job better and what vision testing looks like in your work. How about you start by explaining what you do, why it works, and what problems you may face while doing your job.

Lisa: Hi, sure. Happy to help — though I'll warn you, my job is a lot less interesting than people expect.

So, I'm an operations supervisor at one of our state's driver licensing offices. Mid-sized city, we're one of three branches in the county and the busiest of the three. I've been with the agency fourteen years — started as a customer service rep at the counter, which I think matters, because everything I supervise I've done myself. Six years as supervisor now.

Day to day, I'm responsible for the floor. Staffing the windows, the queue, making sure transactions get processed correctly, handling anything a rep escalates to me, and answering for our numbers when the regional office asks. On a normal weekday we'll do somewhere between three hundred fifty and four hundred fifty transactions. Mondays and the first few days after a holiday are worse. Summer is brutal because that's when teenagers come in.

The vision piece is woven into the license transactions — new applicants, renewals, anybody with a restriction change. We have two desktop screening units at the counter, the kind you put your forehead against and read the line of letters, and a wall chart in the back room for people who physically can't use the viewer. Wheelchair users, somebody with a neck issue, that sort of thing. The rep running the window administers it, marks pass or fail in the system, and moves on. State standard is 20/40 in at least one eye with correction. Below that and we can't issue. They get a paper vision form, they take it to an optometrist or ophthalmologist, the doctor fills it out and signs it, and they bring it back or mail it in and we clear the record.

Why does it work? Honestly, it works because it's fast and it's the same thing we've done for decades. The screening itself is sixty, ninety seconds. Nobody has to be trained for weeks. And it catches the people who really shouldn't be driving — I've had folks come to the window who genuinely could not read the top line. That happens more than you'd think, usually with older drivers who've been quietly getting worse for years and nobody in their family wanted to have that conversation.

Problems. Where do I start.

The biggest one for me is the failure path, not the test. Somebody waits forty-five minutes, gets to my window, fails the screening, and now I'm the one telling them they're not leaving with a license today. They have to find an eye doctor, which might be three weeks out, pay for an exam, come back, and wait again. People get upset, and I don't blame them. Some of them get upset at my reps, and that's the part I actually hate. I've had people cry at the counter. I've had people yell. We had a man last spring who needed his license for work and he just sat down in the lobby and put his head in his hands. There's nothing I can do for him. The rules are the rules.

Second, consistency. My reps aren't eye doctors. They're reading a result off a machine and checking a box. One rep will let somebody try the line a second time if they think the person's glasses are smudged. Another one won't. I've tried to standardize it, but when the lobby's full and we're down two people, everybody takes shortcuts. And one of my units is probably twelve years old. It still works — I'm not saying it gives bad numbers — but it's old, and when the regional office audits our pass-fail rates and asks why my branch is three points off the branch across the county, I don't have a great answer. All I've got is a checkmark in a field.

Third, the paper. Every one of those vision forms is a manual process on our end. Somebody has to receive it, match it to a record, key it in. They get lost. People mail them to the wrong address. We've had applicants swear up and down they sent it in and we have no record of it.

I'll push back on one thing people assume, though — folks hear "DMV" and "vision test" and figure the vision test is what's slowing the line down. It really isn't. What slows my line down is document verification. Somebody brings the wrong proof of residency, or their name doesn't match their birth certificate because they got married, and that's a fifteen-minute conversation. The vision screening is a blip. Where it costs me time is when it fails, or when somebody elderly takes four or five minutes fumbling with the viewer because they can't get positioned right and they're nervous.

Anything specific you want me to go deeper on?

Interviewer: I'm curious about the eye exams. It sounds like a potential compliance risk if equipment is inconsistent. Have you found there to be any problems with this in the past?

Lisa: Compliance risk — that's the language the regional office uses. I'd say it's a liability question more than a compliance one, and the honest answer is: nothing's blown up, but I'd be lying if I said I slept great about it.

Let me give you the specific things that have actually happened.

Three or four years ago, before I had my current district manager, we had a stretch where one of the units was giving — I don't know the technical term — it was the illumination. The internal light was dimming. It's gradual, so nobody notices it day to day. We only caught it because one of my reps, Denise, who's been here longer than me, said something like "people are struggling more at window two than window three." That's it. That's our quality control. A twenty-year employee's gut feeling. We put in a work order, the vendor tech came out, confirmed the lamp was degraded, and we don't know how long it had been that way. Could have been two months. Could have been eight.

Now — did anybody get wrongly failed in that window? Almost certainly, yes. A borderline person at 20/40 or 20/50, with a dim unit, probably got a fail they'd have passed on a good machine. Those people went and paid eighty, a hundred dollars for an exam they maybe didn't need, took time off work, came back. Nobody ever knew. We don't track that. There's no

mechanism to go back and say "these two hundred people were screened on a bad unit, let's review them."

That's the part that bothers me. The failure mode isn't the dangerous one — a wrongly *failed* person just gets inconvenienced and ends up in front of a real doctor, which is fine, if expensive and annoying. It's the other direction I can't verify at all. If a unit were reading generous, somebody who shouldn't be on the road gets a license from my office, and I would have no idea. We have never, to my knowledge, had an audit designed to catch that.

As for calibration — and this is where I think you're going — there's a maintenance contract. A tech is supposed to come through annually. I can tell you that in fourteen years I have never personally seen documentation of a calibration certificate for our units. I assume it exists at the state level. Nobody's ever asked me for it, and I've never been handed one. We log work orders when something breaks. That's the extent of what I control.

The dispute side: we get a handful a year where somebody insists their vision is fine and we got it wrong. Usually it's somebody who feels embarrassed and is pushing back out of pride. My policy is I'll re-administer it myself, on the other unit, and if they pass on the second one I issue the license. Which — I'll say out loud — is not a great policy. If the two machines disagree, one of them is wrong, and I just picked the answer that makes the customer happy and gets them out of my lobby. But nobody's given me a better rule, and I'm not going to stand there and argue with a man about his own eyes when I don't have anything to back me up except a checkmark.

The one that would actually be a problem is a crash. If somebody we screened and passed causes a serious accident, and the attorney pulls our records and asks how we verified that result — what I have is a clerk's checkbox, a date, and an employee ID. No image, no raw acuity score in most cases, no record of how many attempts they got, no record of whether that unit had been serviced. That hasn't happened to me. It's happened elsewhere in the state and I heard about it secondhand, so I don't want to represent details I don't know. But it's the scenario that's in the back of my head.

I'll add one thing, because I don't want to overstate this: I'm not sitting here thinking our office is out of compliance. We follow the statute as written. The statute tells us what standard to apply and that we have to screen. It doesn't say much about proving we screened correctly. So we're compliant with a rule that I think is thinner than people assume it is.

Interviewer: Oh, that's interesting. I only thought there would be wrong diagnoses for patients who shouldn't pass, not the other way around.

I see your standpoint about your office meeting compliance. However, it sounds like your office is set up for failure on that end. Bad process can cause bad results inherently, regardless of the workers.

Something I want to dig into. How much time does a vision test take during an appointment? Theoretically, could it help speed up your wait times if there was a way to verify eye exams prior to coming in for the license appointment?

Lisa: I'll take the first part, but I want to be careful about how I agree with it. "Set up for failure" is strong. I've got a process that was designed in, I don't know, the eighties, for a world where this was the best you could do at a counter, and it hasn't been rethought since. That's not the same as broken. It catches the people it was built to catch. What it doesn't do is produce evidence, and I think the gap between "we did it right" and "we can show we did it right" is wider than anybody above me has had to look at. I'd put it that way.

On timing. The screening itself, if everything goes smoothly, is under a minute. Call it forty-five seconds to a minute and a half depending on who's at the window and whether they're wearing their glasses. For an older applicant, somebody anxious, somebody who doesn't speak English well and my rep is working through it — three, four, five minutes. I've seen longer.

But I want to be precise, because I think your question has an assumption baked into it. The screening is not what makes people wait. People wait because there are forty of them and six windows. The screening is a minute out of a transaction that runs ten to twenty minutes. If you deleted the vision test from my process entirely — just removed it — I'd save maybe four or five percent of counter time. My wait times would not meaningfully change. Document verification, payment processing, photo capture, people who show up without the right paperwork, that's the mass of it.

Where it costs me real time is the failure. That's not a minute, that's the whole transaction collapsing. Rep has to stop, explain, print the form, explain again because the person's upset, handle the emotional part, and now we're eight or ten minutes deep and nothing got issued. Then that person comes back a few weeks later and consumes a second full slot. So one failed screening costs me roughly two transactions plus the overhead. Across a year that's real. And the lobby feels it, because that window is tied up and everybody can see it.

So on your actual question — could verifying an exam beforehand help? Let me separate two things, because I think they're different problems.

If you mean a way for someone to find out *before they come in* whether they're likely to pass, so the people who'd fail go see a doctor first and show up with the form in hand — yes. That's useful to me. Not because of the minute at the window. Because those are my worst transactions and my angriest customers, and because that person's second trip is a slot I could have given to somebody else. If even half the people who currently fail at my counter arrived already cleared, I'd notice that in my day.

If you mean replacing my in-office screening with some result the person did at home — that's a different conversation and I'd be a lot more cautious. Two reasons, and neither is about the technology.

First is statute. The law says the applicant is screened at the point of application. That's not my rule to bend. Changing it means the legislature or at minimum a policy directive from the department, legal review, and that's years, not months.

Second is identity and integrity, and I'd push on this hard. At my counter, I have the person's face, their documents, and my rep watching them. I know who took the test. If it happens at home, I don't know whether that's the applicant or their nephew with better eyes. I don't know if they had a chart taped to the wall. I've had people memorize my wall chart — the letters don't change, people talk, and the same families come through. If I can't put a face to the result, it's not evidence to me, it's a claim.

Now, the doctor's form I already accept is also just a piece of paper somebody hands me, and I'll grant you I don't verify those very hard either. A signature and a license number. So I'm not pretending my current bar is high. But there's a name and a license on the line, and that's a person who loses something if they lie. A website doesn't lose anything.

One more thing you didn't ask. If something gave me a result I could actually see at the counter — not replacing my screening, just telling me "this person was checked by Dr. So-and-so on this date, here's the acuity" — the thing I'd want most out of it isn't speed. It's that it would be the first time I had a number in the record instead of a checkbox.

Interviewer: I see. So at the end of the day, there has to be significant verification if someone were to complete the exam before coming in. That way it can be verified that the person obtaining the license actually took the exam themselves.

I want to dig into that a little deeper; for individuals with commercial drivers licenses, how often do they need to have their vision checked? I'd think it's important for individuals driving at all, including non-commercial drivers, to have their eyesight checked on a more frequent manner to maintain public safety. What's your take on that?

Lisa: Let me be upfront about the CDL part, because I don't want to give you a confident wrong answer on something that's federally regulated.

Commercial is a different animal and it's not really my lane. CDL medical certification runs through the federal side — the medical examiner's certificate, the DOT physical. It's a certified medical examiner, not a counter clerk, and the vision standard is stricter than ours. There's a peripheral field requirement and a color recognition piece that we don't test for at all on the regular side. My understanding is the certificate runs up to two years and can be shorter if the examiner flags something. Our role at the branch is mostly that we need a current certificate on file to keep the commercial privilege active, and if it lapses, the system downgrades them. If you need exact intervals and thresholds you should pull the federal regulation, because I'd be going off memory and the numbers have moved around.

But the shape of your question is the right one. There's a commercial driver who has to sit in front of an actual medical examiner every couple of years, and then there's a sixty-eight-year-old in a Buick who I screened once, for ninety seconds, eight years ago.

Our renewal cycle is eight years. Some of those renewals can be done online, and when they're done online there's no screening at all — it's an attestation. The person checks a box saying their vision hasn't materially changed. So in practice there are people on my roads who have not had their eyes looked at by anybody, in any capacity, in sixteen years, because they renewed online once in the middle. We do require an in-person renewal periodically and there are tighter rules for older drivers, but I'd want you to check the specific ages against our state's rules rather than take my word for it.

Now you asked for my take, so here it is, and I'll flag that this is Lisa's opinion from the counter, not the department's position.

Yes, I think eight years is too long. I see the evidence of it. The people I fail are almost never a surprise to me — by the time they're at my window squinting, their vision has been going for years, and usually somebody in their life knew. Daughter brings Dad in, and she's standing behind him making a face at me, because she's been trying to have this conversation for two years and she's hoping I'll be the bad guy so she doesn't have to be. That happens constantly. I've become a tool families use to take car keys away, and I don't love that, but I understand it.

Here's where I'd complicate it, though. I'm not sure more frequent *screening* is the answer, and I say that as the person who'd have to do the screening.

If you told me tomorrow that every driver needs a vision check every two years instead of eight, you've just quadrupled the volume coming through my lobby, and nobody is quadrupling my staff. My wait times go from forty minutes to a couple of hours, people start skipping renewals or driving on expired licenses, and now I've made a different public safety problem. That's not a hypothetical — we saw what happened when the REAL ID deadlines hit. The lobby doesn't absorb it.

And the screening we do is genuinely crude. It's acuity, one distance, one condition, good lighting, no glare. The thing that scares me about older drivers isn't whether they can read a line of letters in a lit box at the counter. It's night driving, glare recovery, peripheral loss, cataracts, somebody whose central vision tests fine but who has a hole in their field they've learned to compensate for. I can't see any of that. Somebody could pass my screening cleanly and be genuinely unsafe on a dark road, and I'd issue the license and have no idea. So I'm cautious about the idea that running my test more often buys as much safety as people assume. It might just mean I'm running a weak test more frequently and generating more paperwork.

What I actually think — and this is me talking, not policy — is that the problem isn't the interval, it's that there's no connection between the eye doctor's chair and our records. A person could have been diagnosed with something significant by their optometrist last month, and when they walk up to my counter I know nothing about it. The only way it reaches me is

if the person tells me, or if a physician reports them, which happens rarely and is an uncomfortable thing for a doctor to do to their own patient. So we rely on a ninety-second test at the counter because it's the only signal we actually have.

If the signal already existed somewhere else, I wouldn't need to be the one generating it.

Interviewer: I understand that about congesting your office with more frequent screening. However, in a world where that theoretically wouldn't happen, it seems that having more frequent screening would be beneficial if it came without consequence.

Back on the CDL topic. I understand that the medical verification is outside of your domain. However; if vision testing was more accessible and verified, would you want the process to involve non-CDL drivers getting tested more frequently?

Lisa: I want to push on the framing a little, and then I'll answer you straight.

"Without consequence" isn't a world I know how to think in. Everything in my building has a consequence. Even if the test took zero time and cost nobody anything, you've still created a result that has to go somewhere, and now somebody's record says something, and if it says the wrong thing I've got a person at my counter who can't get to work. The failure path doesn't disappear just because the test gets cheap. It gets bigger, because more people are taking it. So when you say more screening is beneficial if it's free — more *correct* screening would be. More screening at the accuracy my machines give me, I'm not as sure about. I'd be generating four times as many of those conversations and four times as many of the wrongly-failed people I mentioned earlier, and those are real people spending a hundred dollars and a day off.

That's my caveat. Here's the answer to what you asked.

Yes. If it were accessible, meaning the person doesn't have to take a day off and pay a hundred dollars to get it done, and verified, meaning I know it was that person and the result is trustworthy — then yes, I'd want non-commercial drivers checked more often than every eight years. Eight years is indefensible and I don't think anyone at the department would defend it on the merits. It's a number that exists because renewals cost money to process and the state stretched the cycle to save on operations. That's it. There's no safety reasoning behind eight.

But let me tell you what I'd actually want, because I don't think "more often" is quite it.

I'd want something closer to how the commercial side works in structure, not in burden. On the commercial side there's a qualified person doing the examination and a certificate that lands in the record, and the agency isn't the one performing the exam — we're the one holding the result. That division makes sense to me. The agency is not a clinical operation and shouldn't be. My reps are good at verifying documents and processing transactions. They are not and should not be the state's vision screening infrastructure. Right now they are, by default, because nobody built anything better.

So if you're asking me to design it from my seat: I'd rather the check happen out in the world, by someone accountable, and have it flow into the record, than have it happen at my counter more often. Then the renewal interval question gets easier — I wouldn't need the person to come in to be screened, I'd need the record to show a current result. Someone who sees their optometrist every year is already generating that data and it's just evaporating.

Two things I'd hold firm on, though.

The verification has to be real, and I've said that already, but I'll put a finer point on it. It isn't enough that the test is accurate. The chain from the person to the result has to be something I could defend to a lawyer. Who took it, when, under what conditions, who stands behind it. If all I get is a better number with the same weak chain, I've gained nothing for the scenario that actually worries me.

And there'd have to be something in it for the driver, not just for the state. Right now I'm the enforcement point and that's all I am — I'm where you find out you've lost something. If this just becomes a more frequent, more accurate way for people to lose their license, people will avoid it the way they avoid us now. Somebody at my counter is not thinking about public safety. They're thinking about getting to work tomorrow.

Can I ask what's prompting the CDL comparison? I want to make sure I'm actually answering what you're after, because commercial really is a different world and I'd hate for you to build something on my secondhand understanding of it.

Interviewer: That makes way more sense. I think that you all maintaining a copy of the certificate is beneficial from a liability perspective. It lets you offload the liability risk but still have verification that an individual qualified for their license. All while maintaining a compliant transfer mechanism that preserves the validity. That's an interesting perspective.

The CDL comparison is mostly my own interest. This job is very unfamiliar to me, so I'm trying to understand if there are comparisons that are worth noting, or if these really should be independent of each other entirely. From my perspective there are definitely similarities, but I don't know what I don't know. I'm trying to gather more context.

Lisa: Fair enough, and I appreciate you saying that, because people usually come in here already knowing what they think.

Let me try to give you the honest map, including the parts where I'm guessing.

Where they're genuinely similar: both are a government body deciding whether a person meets a physical standard to operate a vehicle. Both use acuity as the main proxy. Both have the same structural weakness, which is that a person's eyes change continuously and we check discretely, so there's always a gap between what the record says and what's true. And both depend on somebody being honest — the commercial driver who doesn't mention his diabetes to the examiner is the same problem as the guy at my counter who's memorized the chart.

Where I'd tell you they're different, and this is the part I think matters most for what you're doing:

The commercial side is an employment system. There's a carrier, and the carrier has insurance, and the insurance has requirements, and there's a safety director whose job depends on their drivers being qualified. The driver has a financial incentive to stay certified because it's his livelihood, and his employer has one too. There's a whole structure around that person pushing in the same direction. Non-commercial has none of that. It's one individual, a state agency, and an eight-year gap. Nobody is checking on that person. Nobody's insurance is asking. That's not a small difference — it's basically the whole difference, and I'd argue the commercial system works because of that surrounding structure more than because the standard is stricter.

Second difference: the commercial exam is done by a certified medical examiner, which means there's a licensed person on a registry whose name is attached and who can be removed from that registry. That's the accountability chain I keep coming back to. On my side, the equivalent person is a twenty-three-year-old counter clerk making state wages who got half a day of training. I don't say that to run my people down — Denise is better at reading a person than any machine I've got. But if you're asking where the liability actually sits, it sits on the agency, because the individual who administered it has no license to lose.

Third, and this is the one I'd flag hardest if you're trying to learn something from the comparison: volume and voluntariness. Commercial drivers are a small, identifiable, motivated population who will jump through the hoop because it's their job. Non-commercial is everybody. It's a hundred and eighty million people who view this as a chore. Anything you'd design for the general population has to survive contact with somebody who doesn't want to do it, doesn't think it matters, and will take the path of least resistance. Most things that work for a professional population fall apart at that scale.

Where I don't know what I don't know, and I'd want you to go ask someone else: the actual federal regulatory structure, how the examiner registry really functions in practice, whether the certificates get audited and what happens when one is falsified. I've never worked that side. I've only ever been the office that accepts the certificate and keys it in. I'd be careful treating me as a source on it.

Honestly, if you want the comparison properly, you should talk to a medical examiner who does DOT physicals and a safety director at a trucking company. The examiner sees the clinical side, and the safety director sees what it costs a business to keep a hundred drivers compliant. Between the two of them you'd learn more in an hour than I can give you secondhand.

One correction on what you said back to me, though, because I don't want it going into your notes wrong. You described the certificate as offloading liability. I'd be careful with that. It transfers some of it — the clinical judgment lands on the examiner, and that's real. But the agency is still the one that issued the license. If a driver we licensed causes harm, nobody's going to be satisfied that we had a form on file if the form was obviously stale or obviously

wrong and we didn't look. The form moves where the judgment was made. It doesn't move the decision. I'm still the office that said yes.

Interviewer: The fact that you specifically mentioned somebody memorizing a chart is intriguing to me. I've thought through that exact scenario before and it bothers me to know that an individual could pass without inherently cheating; it potentially allows unqualified drivers.

Thank you for sharing the differences. This really helps a lot with my understanding.

The path of least resistance is a great point. It'd have to be "worth it" to the user for sure, or perhaps more heavily enforced as a requirement which doesn't seem super feasible at scale.

Great point about compliance. Thank you for pointing that out.

I think I have all the information I need. Thanks for going through this with me!

Lisa: Happy to do it. It's not often someone asks me about this part of the job — usually people want to complain about the wait or ask why their registration didn't come in the mail.

Two things before you go, since you asked earlier if there was anything you should've asked but didn't.

On the memorization thing — I'd adjust the picture slightly. Most of the people doing it aren't cheating in the way you'd imagine. It's usually not a plan. It's a sixty-year-old who's been coming to this office for thirty years, who kind of knows what's coming, who guesses at the bottom line the way you'd guess on a test, and who honestly believes their eyes are fine. They're not trying to get one over on the state. They're avoiding a conversation they don't want to have about getting older. The wall chart is the worst for it because the letters never change, but even the machines have a limited set. If you're building something around that, I'd think less about catching cheaters and more about the fact that the person taking the test usually doesn't want the answer. That shapes everything.

The other thing you didn't ask about, and nobody ever does, is who else is in the room. You asked who'd need to be involved in a change — I gave you leadership, IT, legal, procurement. What I left out is my reps. Every time the state rolls something new out to us, it arrives as a memo and a half-hour of training, and then the people at the windows have to make it work at eleven in the morning with forty people in the lobby. If it adds a step, they'll find a way around it, and I'll find out six months later. Whatever you're thinking about, the counter clerk is the one who decides whether it actually happens.

And if it ever gets real — nothing changes in a state agency without a budget line and somebody above me deciding it's their problem to solve. Good luck with that part. That's the hard one.

Good luck with your project.

Burak Donbekci – Anxious Applicant Alex

Interviewer: Tell me a bit about yourself and how you think about your eyesight day to day.

Alex: Sure. I'm 20, I'm a junior in college, and I'm studying business. Most of my day is classes, my laptop, and my phone. I don't wear glasses and never have. Both my parents do, though. My dad got his in his twenties and my mom wears progressives now, so it's kind of always been in the back of my head that I'm probably next.

Day to day I don't really think about my eyes unless something reminds me. Like, I'll be in a big lecture hall and realize I'm squinting at the slides from the back row, so I'll move up a few rows. Or I'll be out with friends and someone will read a menu board from across the room and I'll pretend I can read it too. I can't tell if that's normal or if my eyes are actually getting worse. That's kind of the whole thing for me. I just don't know.

Interviewer: Walk me through the last time you had your vision checked. What prompted it?

Alex: Honestly, I had to think about this one. The last time anyone officially checked was when I got my learner's permit at 16. At the DMV they have you look into that machine and read lines of letters. I passed, but I remember the last line was hard and I kind of guessed on a couple of letters. Nobody said anything, so I figured I was fine.

Before that it was school screenings and sports physicals in high school, where a nurse has you cover one eye and read a chart on the wall. That was a two-minute thing. I don't think I've ever been to an actual eye doctor, except maybe once when I was really little and my mom took me because of my dad's eyes. I don't remember it.

What's prompting this now is that I never finished getting my full license. I've been driving on and off with my permit, and now I have an internship next summer where I'll need a car, so I have to go do the license. And I keep thinking about that vision machine.

Interviewer: How did you get that check done? Where did you go, how long did it take, and what did it cost?

Alex: The DMV one was part of the permit visit. My mom drove me, and we were there for something like two hours total, mostly waiting for our number to be called. The vision part itself was maybe a minute at the counter. It didn't cost anything separate. It was part of the permit fee, which my parents paid.

The school ones were free and took basically no time. You'd get pulled out of class, read the chart, go back.

So I've never actually paid for an eye check myself, and I don't really know what it would cost. I'm still on my parents' insurance, but I genuinely have no idea whether that covers eye stuff. I'd have to ask my mom, and I've sort of been avoiding asking because then it becomes a thing.

Interviewer: What was the most difficult or frustrating part of that experience?

Alex: At 16, nothing really felt difficult, because I didn't know enough to be worried. The frustrating part is more looking back on it now. I guessed on that last line and passed, and that means I don't actually know if I passed because my eyes were fine or because I got lucky.

The DMV itself was the annoying part in general. The waiting, the lady at the counter being kind of short with people, everyone in the room watching whoever's at the window. I remember the guy before me had some issue with his documents and had to leave, and I felt bad for him, because you could tell he'd been waiting as long as we had.

I'll be honest, though, if you're asking what I'm actually nervous about for the license, the vision test isn't the biggest thing. The road test is. Parallel parking with an examiner next to me is way scarier. The vision part is more of a background worry, like one more thing that could go wrong on a day that's already stressful.

Interviewer: How do you currently tell whether your vision has changed between checkups?

Alex: I don't really have checkups, so I just kind of compare to how things used to feel. Like, in high school I could read the board from anywhere, and now I sit closer in lectures. But I also sit in way bigger rooms now, and professors use tiny fonts on slides, so I don't know if that's my eyes or the rooms.

Sometimes I'll test myself with street signs while I'm walking. I'll try to read one, then walk closer and see when it gets clear. Or I'll cover one eye and then the other to see if one is worse. I think my left eye is a little worse than my right, but I'm honestly not sure.

At night it's harder. When I've driven at night with my permit, the signs feel blurrier and headlights have this glare. My dad says that's normal. My mom says I should get checked. So I get two different answers at home, and I don't have a way to settle it.

Interviewer: Have you ever put off getting your vision checked? What made you decide to wait?

Alex: Yeah, I guess I've been putting it off for like a year. Partly it's that nothing is bad enough to force me. I can still see, I'm not walking into things, so it never feels urgent.

But if I'm being honest, part of it is that I don't really want to find out. If I go and they say I need glasses, then I need glasses, and that's a whole thing. Picking them out, paying for them, wearing them. Right now I can kind of live in "maybe I'm fine."

And I don't actually know how to start. I don't have an eye doctor. I'd have to find one near campus, figure out insurance, probably call my mom about the insurance card, and book something in between classes. None of that is hard exactly, it's just a lot of small steps, and there's always something more pressing, like an exam or a deadline. I'm not sure I'd say the steps are what's stopping me, though. If I really thought something was wrong, I'd figure it out. It's more that I'm not convinced it's worth it yet.

Interviewer: Have you ever tried checking your vision on your own, online or otherwise? What was that like, and did you trust the results?

Alex: Yeah, actually. A few weeks ago, kind of late at night, I was stressing about the license and searched "DMV vision test practice." I found a couple of sites with eye charts. One had me stand back from my laptop and read letters, and another one was on my phone.

I didn't really trust either. The instructions said to stand a certain distance away, but I was in my dorm room, so I just guessed. My laptop and my phone are different sizes, so I didn't know if the letters were the right size. I did one test twice and got different results, which made me trust it even less. One of the sites had a big "buy glasses" button, which made me think the whole point was to sell me something.

The bigger thing is that it didn't tell me the one thing I actually wanted to know, which is whether I'd pass the DMV one. It gave me something like "20/30," and I didn't even know if that's a pass or a fail. I had to google what the DMV standard is, and the first answers were for a different state. So I closed it and felt maybe more nervous than before.

Interviewer: What would a vision result need to have for you to trust it enough to act on it?

Alex: I think first it would have to be clear about what it's checking against. Like, "this is the standard your state uses, and here's whether you're above or below it." A random number doesn't help me.

Second, it would need to be consistent. If I do it twice and get two different answers, I'm done with it. And I'd want to know I did it right, like where to stand and whether my lighting is okay.

Third, I'd want it to come from somewhere real. A doctor, a hospital, or the DMV itself, not some site with ads all over it.

And honestly, I'd want my parents to believe it too. If I showed my mom and she said "that doesn't count, go see a doctor," then it wouldn't really settle anything. So it'd have to be legit enough that it ends the conversation at home, not starts a new one.

I'll also say, if it told me I'd probably fail, I don't think I'd do anything right away. I'd probably freak out a little and then sit on it for a while. It would take a bit for me to actually book an appointment.

Interviewer: What happens if a vision problem goes unnoticed for a while?

Alex: For me, I think it would mostly show up in small ways first. Sitting closer in lectures, missing stuff on slides, maybe getting headaches I'd blame on not sleeping. I'd probably just adjust and not even connect it to my eyes.

The one that actually worries me is driving, especially at night. If my eyes are worse than I think and I'm driving in the dark on roads I don't know well, that's a real risk, and not just to me. That one I think about more now that I'm about to be driving for real.

And then there's the DMV angle. If I go in and fail the vision part, I'm the person getting sent away from the counter, having to go find an eye doctor, and coming back another day. I'd be

annoyed about losing the time more than anything. Embarrassed a little too, but mostly it would just mess up my plans, because I need the license before summer.

Interviewer: Is there anything about this topic I should have asked but didn't?

Alex: Hmm. You didn't really ask about my parents. They're kind of a big part of this for me. My mom brings up getting my eyes checked pretty much every time I'm home, because of the family history. It's turned into one of those topics I avoid. So a lot of my not going is mixed up with not wanting to admit she might be right.

You also didn't ask what I know about where to go. I actually don't know if campus health does eye stuff. Maybe they do and I've just never looked. I think a lot of people my age don't know where you'd even start, since our parents always handled doctor appointments.

And one more thing. You didn't ask about cost. I'm a college student, so if checking my eyes costs real money, that's going to push it down the list even more. I'd want to know upfront what it would cost before I started anything.

Interviewer: That's the end of the interview.

Alex: Cool, thanks. Good luck with the project.

In-person interviews (with real people)

Paste your notes or transcript from each in-person interview. Don't reveal your problem statement before the interview — keep it spontaneous.

Paste all in-person interview notes here. Label each: [Team member name] — [Interviewee].

Haydn Stucker – Anna Altmann (notes, not word for word transcript)

Anna meets the full definition of Persona 1 to a T. She is 25, works an office job with most of her day on a computer, and has worn glasses since she was a kid. She has an annual appointment with an optometrist to adjust her lens prescription, and wears both contacts and glasses (primarily glasses, contacts are reserved for active environments, like sports).

Primary Observations:

1. Anna typically starts to notice that her prescription is starting to fall out of focus months before her annual eye doctor appointment. Usually, it is between 2-4 months beforehand, but she deems it's not worth rescheduling and just "waiting it out".
2. When Anna does this, she starts to notice eye strain more often during her workday. Having slightly out of focus glasses, paired with being on a computer screen most of the day, starts to make an impact on her eyes. She has tried blue light glasses with contacts before, but this does not help with the strain; that leads me to believe the strain is primarily from out of focus lenses and not an external cause.
3. Anna's prescription, when it does change, usually changes very little; changes that she finds frustrating that she needs to schedule an appointment to find out. She said that she would prefer an option where she could quickly take a test when she feels

her prescription starting to slip but finds it more difficult than it's worth to schedule an eye appointment.
4. Anna's biggest complaint is that "it shouldn't take so much time to find out I need slightly new glasses, when nothing else is wrong". The inconvenience of how optometrist offices are run is enough to keep her from getting proper care because it is too long of a process to make a meaningful difference.
5. The recurring theme that I (the interviewer) noticed the most is inconvenience. Most local optometrist offices are staffed or managed in a way to efficiently serve their patients when they need new prescriptions. Finding a simpler, more convenient solution to complement a doctor would change this interviewee's approach on getting proper eye care, it seems.

Burak — Kevin W. (akin to persona 4)

1. 23, new grad, started a full-time job this month, no glasses. His mom, dad and sister all wear them; he jokes he "dodged the family curse." Sees his eyes as one of the few parts of his health he's never had to deal with.
2. Has seen an eye doctor once, around age 12. The optometrist said he was borderline and suggested glasses for reading the board. Since then, the only check was the DMV at 19, which he passed with minimal trouble. He can't recall if the issue was eyesight or sign knowledge.
3. The childhood exam was at a local optometrist, and his parents handled the cost. The DMV vision check was quick and forgettable. Now has his own insurance through work and assumes vision is included, but skipped that section of the benefits packet during onboarding.
4. Most frustrating was the childhood glasses: he wore them about a month, noticed no difference, lost them, and never followed up. It left him feeling the whole thing was pointless. At the DMV, the annoyance was purely logistical: he failed the road test three times and had to rebook weeks out each time.
5. Doesn't track it. On two monitors all day, he has bumped up the font size and turned on phone zoom, but sees that as normal screen habits, not a sign of anything. Notices tired eyes by Friday and blames long weeks and hours.
6. Hasn't been to an eye doctor in over ten years and doesn't see that as "putting it off," because nothing has prompted it. "Don't fix it if it ain't broke." Cost and insurance aren't barriers anymore. He simply has no reason to go.
7. Has never tried an online or at-home vision test and doesn't see the point. Tried on his sister's and others' glasses for fun; everything looked worse, which he tends to take as proof his own eyes are fine.
8. Credibility isn't the issue for him. He'd believe most results. The issue is effort. A check would have to be folded into something he's already doing, like an annual physical. A standalone app or website "wouldn't even register." If it takes a separate step, he won't take it.
9. Thinks the stakes are low at his age, since plenty of people go years with slightly off vision. Has never connected his road test failures to his eyes and doesn't think there's a link. At most expects some extra eye strain at work.

Hadley – Anna K.

Anna fits Persona 1 (Routine-Check Riley) except at an earlier life stage. She is about 24, a grad student who spends most of the day on a laptop and has worn glasses since 4th grade. She owns contacts but rarely wears them because her eyes dry out by the end of the day.

Primary Observations:

1. Her last eye exam was about a year ago before moving for grad school. She only went because her mom booked it and she had bent her glasses. She described the exam as easy and had few complaints (only dilation and a 30ish minute wait), mainly because her parent handled the scheduling and payment.
2. She has recently started sitting closer to the front of lectures to read the slides but hadn't connected this to her vision until asked. She doesn't track changes and doesn't know her prescription.
3. She is actively putting off a checkup. She doesn't know which local eye doctors take her parents' insurance or whether it works out of state, so she plans to wait until winter break and go home. She basically said “I can still see fine with my glasses, it's just not perfect."
4. She didn't know online vision tests existed and would be skeptical of one, saying that If it's just an app telling her her vision is fine, she'd be like, “okay, says who." She would trust a result if a real eye doctor reviewed it or friends recommended it, but wouldn't buy glasses based on an app alone.
5. Her biggest concern was insurance, which we had not hypothesized. She says once she's off her parents' plan she'll probably just avoid going because she don't want to deal with figuring it out.

The recurring theme I noticed was dependence on someone else (a parent, a familiar doctor) to make eye care happen. Without that support, uncertainty about where to go and how insurance works becomes the main barrier, more than time or cost. This suggests young adults who have just moved or are coming off a parent's plan could be a possible early-adopter group.

Virtual interviews (with the AI)

Paste each AI persona transcript below. Tip: you can ask the AI to output the full transcript so far, then paste it here.

Paste all virtual interview transcripts here. Label each: [Team member name] — [Persona name].

Fan – Parent Pat

1.Tell me a bit about yourself and how you think about your (or your child’s) eyesight day to day.

I work full time and have two children, one in preschool and one in elementary school. I don't think about their eyesight every day unless I notice something unusual. My older child sometimes sits close to the television, but I can't tell whether that's a habit or a sign that something is wrong. I mostly rely on their pediatrician and school screenings to flag problems.

2. Walk me through the last time you (or your child) had your vision checked. What prompted it?

My older child had a school screening and brought home a letter recommending a follow-up exam. That surprised me because they hadn't complained about trouble seeing. I called the pediatrician's office to ask what the letter meant, and they suggested scheduling an appointment with an eye doctor.

3. How did you get that check done? Where did you go, how long did it take, and what did it cost?

I found an eye doctor who sees children through our insurance directory. The first appointment that worked with our schedule was several weeks away. The visit took about an hour, but with travel and picking my child up from school, it used most of the afternoon. We paid a copay. I don't remember the exact amount, but figuring out which insurance coverage applied took a couple of phone calls.

4.What was the most difficult or frustrating part of that experience?

The uncertainty while we waited. I didn't know whether the school letter meant something serious or whether my child had simply struggled with the screening. Scheduling was also difficult because I had to arrange time off work. The exam itself was manageable once we got there.

5.How do you currently tell whether your (or your child's) vision has changed between checkups?

Mostly by watching their behavior. I notice whether they squint, hold books very close, or complain about headaches. Sometimes I ask their teacher whether they seem to have trouble seeing things in class. But those signs are hard to interpret. Sitting close to a screen doesn't necessarily tell me that their eyesight has changed.

6.Have you ever put off getting your vision checked? What made you decide to wait?

Yes. Before the school letter, I had wondered about the television habit but hadn't booked an appointment. My child said they could see fine, and the previous screening hadn't raised any concerns. With work and other appointments, I kept putting it off. Once we received a specific recommendation, I made the appointment.

7.Have you ever tried checking your vision on your own, online or otherwise? What was that like, and did you trust the results?

I once pulled up an eye chart online and asked my older child to read it. I wasn't sure how far away they should stand or whether the chart was the right size on our screen. My younger child treated it like a guessing game. I didn't trust the results enough to decide whether either of them needed an appointment.

8.What would a vision result need to have for you to trust it enough to act on it?

I would want to know that the test was appropriate for my child's age and that we had done it correctly. I would also want a clear explanation of what the result meant. If the pediatrician or eye doctor recommended it, I would feel more comfortable using it. One reassuring result wouldn't make me ignore a concern raised by their teacher.

9.What happens if a vision problem goes unnoticed for a while?

I worry that a child could struggle at school without knowing how to explain the problem. Adults might think they aren't paying attention when they actually can't see clearly. At the same time, I don't want to assume every reading difficulty or headache is caused by eyesight.

15.Is there anything about this topic I should have asked but didn't?

How differently the two children respond to testing. My older child can explain what they see, but my younger child gets distracted and sometimes guesses just to finish. Also, I don't necessarily want to check their vision constantly. I want to know when there is a reasonable reason to get it checked.

In-person interviews (with real people)

Paste your notes or transcript from each in-person interview. Don't reveal your problem statement before the interview — keep it spontaneous.

Paste all in-person interview notes here. Label each: [Team member name] — [Interviewee].

Fan – My Mother

1. Tell me a bit about yourself and how you think about your (or your child's) eyesight day to day.
I have two children, and I'm quite concerned about their eyesight because nearsightedness runs in our family. I worry that they may develop it if we don't take care of their eyes, so I want to do whatever I can to prevent it or slow its progression.

2. Walk me through the last time you (or your child) had your vision checked. What prompted it?
We take them for a comprehensive eye examination every summer to monitor changes in their vision.

3. How did you get that check done? Where did you go, how long did it take, and what did it cost?
I take them to a local eye clinic. The full examination usually takes about two hours. Their annual exam includes corneal topography, intraocular pressure measurements, a fundus examination, and axial length measurements to monitor myopia progression. Only a few clinics in our area offer all of these tests. The examination is covered by our health insurance.

4. What was the most difficult or frustrating part of that experience?
Probably getting an appointment, because the wait can be quite long. The most frustrating part is finding out that their nearsightedness has worsened and that they need a stronger prescription. I would prefer more frequent checkups so we could understand how their eyesight is changing and decide when to take action. However, additional visits would cost more and be inconvenient.

5. How do you currently tell whether your (or your child's) vision has changed between checkups?
I keep all their examination records, and the clinic also has records of their previous visits. I use those records to compare the results over time.

6. Have you ever put off getting your vision checked? What made you decide to wait?
Yes, we have delayed checkups occasionally, but we still make sure they have at least one examination each year.

7. Have you ever tried checking your vision on your own, online or otherwise? What was that like, and did you trust the results?
Yes, we have an eye chart at home. However, kids tend to memorize the letters or symbols, so eventually it becomes less useful for checking their vision.

8. What would a vision result need to have for you to trust it enough to act on it?
It would need to be accurate and consistent.

9. What happens if a vision problem goes unnoticed for a while?
Their prescription can become significantly stronger within just six months, and the children may not notice the change themselves. I worry that if they continue wearing glasses that are no longer strong enough, their nearsightedness could progress even faster. That is why I would prefer checkups about every three months, especially during their teenage years.

Virtual interviews (with the AI)

Paste each AI persona transcript below. Tip: you can ask the AI to output the full transcript so far, then paste it here.

Qingyi Hu—Licensing Supervisor Lisa

10. Sure. When an applicant comes in for a driver's license renewal or a new license, they start at the front counter with one of our customer service representatives. As part of the application, the staff member has them look into a desktop vision screener machine—the kind with the lenses where you read lines of letters or identify lights for peripheral vision—or in some smaller branch stations, a standard wall chart. The clerk records whether they passed or failed in our system. If they pass, they move on to photo processing or payment. If they fail, or if they have certain medical conditions, we can't issue or renew the license right then. We hand them a physical paper form (a vision evaluation form) and tell them they need to take it to an optometrist or ophthalmologist to get a full examination. Once the doctor fills out and signs that paper form, the applicant has to come back to our office or mail/fax it back to us so we can process it manually and clear their record.
11. Plenty. First, from a line-management perspective, vision screening is a bottleneck. If a machine acts up, or if an applicant is elderly and takes extra time adjusting to the viewer, the counter clerk gets backed up, and the waiting room queue stalls. Second, for customers, failing the in-office test is a huge headache. They've already waited in line for an hour, only to be turned away and told to make an eye doctor appointment, which can take weeks, and then come back and wait in line again. Third, for staff and compliance, our clerks aren't trained opticians. They just read what the machine shows or what the applicant says. Sometimes applicants try to memorize the charts or squint/guess, and clerks don't always catch it. Also, handling paper medical forms leaves room for lost paperwork or manual data entry errors when clearing a driver.
12. To be blunt, we rely heavily on the machine calibration and basic staff training, but consistency is definitely a weak spot. Different counter clerks might administer the test slightly differently—some give applicants a second or third chance to read a line, while others stick strictly to a single attempt. Equipment wear and tear is another issue; some of our vision screening units are over ten years old. When an applicant fails and insists their vision is fine, or when an audit flags inconsistent pass rates across different branches, we don't have a video record or dynamic re-test data to point to. All we have is the clerk's checkmark. If a result is formally disputed or involves a driver license suspension, we have to rely entirely on the outside eye doctor's signed paper report as the source of truth.
13. Time and public trust are our biggest costs. Every extra minute spent on a prolonged or retried vision test adds up across hundreds of applicants a day, increasing wait times and public frustration, which leads to complaints to state representatives. In terms of risk, the stakes are very high. If a driver with inadequate vision passes because they memorized a chart or because a machine wasn't calibrated correctly, that's a serious public safety hazard on the roads. On the flip side, if we wrongly fail someone due to bad lighting or poor machine instructions, we've wasted that citizen's time and money forcing them to pay out-of-pocket for an unnecessary clinical visit.
14. It's a multi-layered process because we're a state government agency. First, our State Department of Motor Vehicles leadership and policy directors would need to approve any new technology or protocol. Second, state IT and cybersecurity teams would have to vet any digital platform for compliance with state data security and privacy regulations (like handling PII). Third, our legal team would need to ensure the new method complies strictly with state motor vehicle statutes regarding vision standards (like 20/40 visual acuity thresholds). And finally, procurement and budget officers would be involved, because government purchasing cycles are notoriously slow and tied to state fiscal budget approvals.
15. You didn't ask about off-site or remote pre-screening possibilities. Right now, everyone assumes vision testing must happen physically inside a DMV branch. But if applicants could reliably complete a standardized, tamper-proof vision screening *before* coming into the office—or renew their license online with a verified vision check—it would take massive pressure off our branch locations. However, the key challenge there is integrity and verification: how do we guarantee that the person taking the test at home is actually the driver, and that they aren't

cheating or using assistance? If a solution solves that verification challenge for state agencies, it would be a complete game-changer.

In-person interviews (with real people)

Qingyi Hu—Individual Linda

1. The interviewee wears glasses and notices that her vision changes during the day. Her eyes can feel blurry shortly after waking up, improve after washing her face, and become less clear again at night, especially when using her phone in a dark environment.
2. Her most recent vision check was about one year ago. Her prescription had increased slightly, and she replaced her glasses. She said that one reason for getting them checked was that her old glasses/frame had been used for a long time.
3. She went to an optical store for the vision test. She did not identify money as the main issue. Instead, the main cost was the time and effort involved in making the trip. She described herself as being reluctant to go rather than saying the test itself was too expensive.
4. The biggest difficulty was the inconvenience of having to go somewhere specifically for the test. She said she was often simply “too lazy to go.” There was also an emotional barrier: she sometimes did not want to test her vision because she was afraid that her prescription might have increased.
5. She mainly relies on her own perception. For example, she notices when things become harder to see clearly, particularly when using her phone at night or in dark environments.
6. Yes. She mentioned that there had previously been a period of around five or six years when she did not get her vision checked. Two reasons emerged: the inconvenience of going for a test and anxiety about finding out that her prescription had worsened.
7. She has not personally used a vision-testing app. However, she has seen image-based vision tests on Xiaohongshu. She did not consider those tests very accurate and therefore did not trust them very much.
8. She said that some form of credible medical or hospital-related certification would increase her trust. She also specifically wanted to know the measurement error compared with the equipment used in hospitals.
9. She believed that the prescription could potentially increase significantly if worsening vision was left unaddressed. Her answer was tentative—she responded along the lines of “the prescription might suddenly increase a lot?”—so this should be treated as her perception rather than a strongly held or medically verified belief.

15. NA

Part 3 — AI Analysis

THE AI WRITES THIS — leave blank until you ask the AI to analyze

HOW TO RUN THE ANALYSIS: (1) Make sure every virtual transcript and every in-person note is pasted into the “Interview Records” section above. (2) SAVE this file. (3) Start a fresh chat with

your AI and UPLOAD this saved file. (4) Send this message: “Read the interview records in this workbook, step out of character, and analyze them as 3A, 3B, and 3C from the AI Analysis section.” Then paste the AI's response into the three boxes below. Re-uploading is required because the AI cannot see interviews from other chats, other team members, or your in-person notes unless they are in the file you upload.

3A. Virtual interview analysis

Key themes & patterns · main pain points · expressed needs & desired outcomes · notable quotes · implications for the hypothesis.

AI fills this in.

Across the virtual interviews, several consistent themes emerged. First, **time, scheduling, and inconvenience** were important barriers to professional vision testing. Routine-Check Riley described multiple small frictions—finding an in-network provider, waiting for an appointment, taking time away from work, and dealing with uncertain costs—that accumulated until the exam was repeatedly postponed. Riley also questioned whether mild symptoms were serious enough to justify a visit.

A second theme was **uncertainty between professional checks**. Riley and Alex relied mainly on subjective signs such as eye strain, blurry road signs, sitting closer to screens, or comparing one eye with the other. Alex’s interview also showed that uncertainty does not always create enough urgency to act; the DMV vision test was only a secondary concern compared with the road test, challenging the assumption that vision screening itself would be a major source of anxiety.

A third theme was **low trust in informal online tests and a need for actionable results**. Riley wanted to know who was behind a test, whether it had been performed correctly, how confident the result was, and what action should follow. A generic result such as “within normal range” was not enough; Riley wanted to know whether professional care was actually necessary. Alex had a similar problem: uncertain viewing distance and different device sizes produced inconsistent results, and a numerical score did not clearly tell him whether he would meet the relevant DMV standard.

The Parent Pat interview added a different need. Parents may struggle to interpret whether behaviors such as squinting, sitting close to screens, or headaches actually indicate a vision problem. The parent also experienced uncertainty while waiting for follow-up care and had to arrange time away from work.

The licensing-supervisor interviews added important organizational nuance. They supported concerns about **inconsistent administration, aging equipment, weak documentation, paper-based referrals, identity verification, and auditability**, but they challenged the assumption that the vision test itself is a major cause of DMV wait times. In the more detailed Lisa interview, a normal screening took roughly one minute and was described as only a small part of the overall transaction. The larger burden occurred when someone failed: staff had to explain the result, issue paperwork, manage an upset applicant, and process a second visit later.

Another institutional theme was that **accuracy alone is insufficient**. A result also needs a defensible chain linking the applicant to the test—who took it, when, under what conditions, and who stands behind the result. Lisa also emphasized that remote testing raises identity and anti-cheating concerns, while laws and policy may still require screening at the point of application.

Notable virtual findings: Riley described the problem as “a bunch of small ones that stack up.” Alex preferred to remain in “maybe I’m fine” rather than confirm that glasses were needed. Lisa challenged the assumption that more frequent screening is automatically better: repeating a weak test more often may create more false failures and administrative burden rather than proportionally improving safety.

Implications for the hypotheses: The virtual interviews supported hypotheses about inconvenience, uncertainty, distrust of informal digital testing, and weaknesses in current licensing workflows. However, some assumptions were narrowed. DMV vision screening itself may not be a major operational bottleneck; the **failure and referral pathway** appears more important. Institutional adoption would also depend on identity verification, traceability, legal requirements, and integration of clinically credible results—not simply greater testing frequency or faster screening.

3B. In-person interview analysis

Same lens as 3A, applied to the real-person interviews.

AI fills this in.

The real-person interviews strongly supported the idea that inconvenience can delay vision care, but they also revealed substantial variation in why people delay. Anna notices prescription changes months before her annual appointment but usually waits because arranging another appointment does not feel worthwhile for a small change. She summarized the problem as: "it shouldn't take so much time to find out I need slightly new glasses." Linda similarly described the main burden as the effort of making the trip rather than the monetary cost, and said she was sometimes simply "too lazy to go." She also revealed an emotional barrier: fear of discovering that her prescription had worsened.

Anna K. described a different, structural barrier. Her last exam was easy because her mother booked it and her parents' insurance paid for it, but after moving for graduate school she did not know which local providers accepted her insurance or whether it applied out of state. Although she had begun squinting at lecture slides, she planned to wait until winter break to see her childhood eye doctor. She explained: "Once I'm off my parents' plan… I'll probably just avoid going because I don't want to deal with figuring it out."

However, Kevin W. challenged the assumption that delayed care is always caused by cost, scheduling, or distrust. He had not seen an eye doctor for more than ten years but did not consider himself to be postponing care; he simply did not feel that anything was wrong. His attitude was essentially "Don't fix it if it ain't broke." Credibility was not his main concern

either. The issue was effort and lack of motivation, and a standalone app or website would not matter enough for him to use it.

The parent interview also produced important nuance. The interviewee already keeps detailed examination records and uses them to compare results over time, so the hypothesis that parents universally struggle to track historical results was not supported. Instead, she wanted more frequent information because annual comprehensive exams were not frequent enough for her preferred level of monitoring. She also found a home eye chart less useful because children could memorize the letters or symbols. In contrast, Anna K. did not track her vision at all and had not connected sitting closer in lectures to a possible vision change until she was asked about it.

Trust in digital or self-administered testing was another important theme, but it varied by person. Linda wanted medical or hospital-related validation and specifically wanted to know the measurement error compared with hospital equipment. Anna K. did not know online vision tests existed and said she would trust a result only if an eye doctor reviewed it or friends recommended it: "If it's just an app telling me 'your vision is fine,' I'd be like, okay, says who." In contrast, Kevin W. said trust was not the issue because he lacked enough motivation to test in the first place. Therefore, trust is important for some users but should not be treated as a universal barrier.

Notable real-person findings: emotional avoidance, lack of urgency, habit, perceived effort, and dependence on others were stronger than expected. Users did not simply weigh cost and accuracy rationally. Some delayed care because the inconvenience felt disproportionate to the expected benefit; others avoided learning bad news; some saw no reason to test at all; and some relied on a parent or familiar provider and stalled when that support was missing.

Implications for the hypotheses: The hypothesis that in-person testing creates time and effort costs was strongly supported. Difficulty monitoring vision was only partly supported because some users rely on subjective symptoms while highly engaged users maintain detailed records. Distrust of online testing was also partly supported rather than universal. The interviews introduced three important additional barriers: emotional avoidance, insufficient perceived urgency, and confusion about insurance and where to go, particularly for young adults who have recently moved or are leaving a parent's care.

3C. Compare and contrast

What was consistent across both? · what differed? · where did virtual interviews miss nuance? · where did virtual interviews add value? · what does this say about each method's strengths and limits?

AI fills this in.

The virtual and in-person interviews were consistent on several broad themes. Both showed that professional testing can involve more effort than users feel is justified, that people often rely on subjective signs between formal checks, and that informal online testing can create questions about accuracy, testing conditions, and what the result actually means. Both also

suggested that a result is more useful when it leads to a clear next action rather than simply presenting a numerical score.

One theme appeared across both methods: young adults depending on parents for eye care. The virtual persona Alex said his parents always handled doctor appointments and that he did not know where to start or whether his insurance covered vision. Anna K. described almost the same situation after moving for school, and Kevin W. had skipped the vision section of his new benefits packet. This suggests that the transition to managing one's own healthcare may be a meaningful trigger point.

The major difference was that the in-person interviews revealed more behavioral and emotional nuance. Virtual personas generally gave structured explanations involving scheduling, insurance, trust, uncertainty, or process design. Real interviewees showed messier behavior: Linda avoided testing partly because she feared bad news, Kevin W. did not perceive a meaningful problem worth solving, and Anna tolerated slightly outdated glasses because another appointment felt disproportionate to the benefit. These findings show that non-use is not always caused by an explicit functional pain point.

The in-person interviews also challenged some broad assumptions from the virtual interviews. Tracking was not universally difficult: one parent already maintained detailed longitudinal records. Trust was not universally the main obstacle either: for one interviewee, effort and lack of perceived need mattered much more. At the same time, the real parent interview reinforced a practical weakness of simple home testing because children may memorize repeated eye-chart symbols.

The virtual interviews added particular value in exploring the institutional licensing context, which was difficult to access through the team's real interviews. The expanded Lisa interview showed that the important organizational problem may not be screening speed itself. Instead, larger concerns included failed-screening workflows, repeat visits, inconsistent administration, weak records, equipment quality, identity verification, legal requirements, and whether a result can be defended later. Lisa specifically distinguished between merely having an accurate result and having evidence of who took the test, under what conditions, and who is accountable for it.

The virtual licensing interviews also demonstrated one limitation of simulation: they can surface plausible operational issues in great depth, but those findings remain hypotheses rather than validated organizational evidence until a real licensing-agency employee is interviewed.

Overall, virtual interviews were useful for rapidly exploring personas, edge cases, institutional workflows, and assumptions that could be challenged, while in-person interviews were stronger for revealing actual behavior, motivation, contradiction, and emotional barriers. Used together, the virtual interviews broadened the problem space, while real interviews helped identify which pain points actually influenced customer behavior.

Part 4 — Team Final Assessment (NO AI)

STUDENT WRITES THIS — DO NOT USE AI HERE
This is YOUR reflection — write it yourselves, not the AI. Synthesize the AI's analysis with what you saw and heard in the interviews.

1. Hypothesis validation

Is your original hypothesis supported, refuted, or partly validated? What evidence backs that?

Your team's response:

What our interview supported

1. In-person vision testing requires time and effort.
2. Parents also face uncertainty between formal checkups. Parents have difficulties interpret whether behaviors such as squinting or sitting close to screens actually indicated a vision problem.
3. Some home/self-tests have practical limitations. In the real parent interview, the children could memorize the eye-chart letters or symbols, which made repeated home testing less useful.

What we missed

1. Emotional barriers can discourage people from getting their vision checked. Some people may avoid testing because they are anxious about finding out that their vision has worsened.
2. People may not trust online vision-testing results. The interviewee was skeptical of informal online vision tests and wanted clinical validation and information about measurement error compared with hospital equipment.
3. Lack of urgency leads to bigger barrier than cost or inconvenience. People who had not seen an eye doctor for years might have vision problems because he simply did not feel anything was wrong; cost and insurance were not the main barriers.

2. Key learnings

What surprised you? Which customer needs did you underestimate or miss? Which assumptions were challenged?

Your team's response:

1. Difficulty tracking vision over time was only partly supported. Some users relied mainly on subjective changes, but one parent already kept detailed examination records and compared results over time.
2. Distrust of online testing was not universal. Several interviewees questioned online-test accuracy, but one real interviewee said credibility was not the issue; effort and lack of motivation mattered more.

3. Vision-test anxiety was weaker than expected for some applicants. Alex said the road test was much more stressful than the DMV vision screening, so the original assumption about vision-test anxiety was only partly supported.
4. Institutional pain points are not yet fully validated. The licensing-agent interview supported concerns about bottlenecks, inconsistent administration, gaming the test, and paper-heavy referrals, but that evidence came from a virtual persona rather than a real licensing employee.

3. Revised understanding

How has your understanding of the customer changed? What would you test or explore next?

Your team's response:

Our understanding changed from viewing the problem mainly as one of cost and convenience to recognizing that customers delay vision testing for several different reasons we initially didn't see coming. Some people want a quicker way to confirm a suspected prescription change, while others avoid testing because they fear bad news or simply do not believe anything is wrong to start with. Trust in digital testing matters for some customers, especially those who want clinical validation and transparent accuracy information, and convenience alone will not motivate everyone.

Parents also have different needs: some struggle to recognize possible symptoms, while highly engaged parents already track examination records and want reliable monitoring between annual visits. Next, we would identify which customer segment has the strongest unmet need and test what level of clinical validation, accuracy information, result explanation, and next-step guidance would make people act. We would also explore whether testing should fit into an existing routine, such as an annual physical, and investigate how device conditions, children's attention, and memorization affect the reliability of at-home testing. Finally, any DMV or licensing use case would need to be validated with real agency employees because the current organizational evidence came only from a virtual interview.

4. Virtual vs. in-person

How did the two interview types compare for uncovering insight? When would you use each in future discovery work? Strengths and limits of each?

Your team's response:

Virtual interviews helped us quickly explore different customer types and identify broad themes such as inconvenience, uncertainty, trust, and organizational challenges. However, their structured answers may have reflected assumptions built into our personas, so they were better for generating hypotheses than validating them.

In-person interviews revealed more realistic behavior and emotional nuance, including fear of bad news, lack of urgency, and resistance to taking an extra step. They also challenged some of our assumptions, although our sample was small.

In future research, we would use virtual interviews early to explore customer segments and improve our questions, followed by in-person interviews to validate findings and understand what customers genuinely experience and value.

5. Next steps

What will you do based on these findings? What questions remain? How does this inform your product?

Your team's response:

The next step is to test three assumptions:

1. what makes users trust a digital vision result
2. how result presentation affects emotional avoidance
3. what kind of guidance makes users take action.

We could create low-fidelity prototypes of different result experiences and compare credibility cues, emotional framing, and next-step guidance. We can also measure which versions increase trust, reduce unnecessary anxiety, and make users more willing to follow the recommended action. Last but not least, we could validate the licensing use case separately with real agency employees before investing in that segment.

Submission & Checklist

Submit one completed workbook per team to Canvas. Confirm you have:

- ☐ Student Inputs complete for each persona (snapshot, JTBD, pain-point hypotheses)
- ☐ Interview questions written
- ☐ All virtual transcripts pasted into Interview Records
- ☐ All in-person notes pasted into Interview Records
- ☐ AI Analysis Parts A, B, and C filled in
- ☐ Team Final Assessment written by the team
- ☐ Each member completed at least one virtual AND one in-person interview

Appendix — Access Rules & Example Personas

Section access legend

Box style	Meaning
Blue	Student writes; AI may read as context but must not edit it.
Green	AI writes it, in the AI Analysis section, only after you upload completed interviews.
Amber	Read it; do not edit it.
Blue with red edge	Student writes with NO AI — your own work only.

> **FOR REFERENCE ONLY**
> These show how to structure a persona. Do NOT copy them into your submission.

Example — "Busy Brenda"

Snapshot

Name	Busy Brenda
Age Range	30–40
Location	Urban metro
Job / Role	Project Manager
Description	Constantly juggling deadlines at work and personal commitments.

Goals & motivations

Primary Goal	Stay organized with minimal effort.
Secondary Goals	Reduce stress; streamline daily tasks.
Motivations	Wants more control and predictability.
Definition of Success	A system that saves time and reduces mental load.

Pain points / challenges

Pain Point	Why it matters
Too many disconnected tools	Causes confusion and wasted time.
Overwhelming notifications	Increases stress and reduces productivity.
Inconsistent task tracking	Leads to missed deadlines.

Example — "Student Sam"

Snapshot

Name	Student Sam
Age Range	18–22
Location	Campus environment
Role	Undergraduate student
Description	Balances classes, study groups, and part-time work.

Goals & motivations

Primary Goal	Stay on top of assignments.
Secondary Goals	Improve grades; manage time better.
Motivations	Wants academic success without burnout.
Definition of Success	A simple system that helps avoid missed deadlines.

Pain points / challenges

Pain Point	Why it matters
Procrastination	Hurts academic performance.
Forgets assignments	Causes stress and grade loss.
Tools feel complex	Leads to avoidance.